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Identifying Young Adults at High Risk for Weight Gain Using Machine Learning

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AUTHOR CONTRIBUTIONS

JAM led the study, prepared the manuscript, and worked closely with our biostatisticians and programmers to analyze the data. JB, LS, and BMH helped with the development and coding of the machine learning models. DJ helped to revise the manuscript and provided significant input for the body of the manuscript. LPH helped revise the manuscript. MMC assisted with the study design, selection of the machine learning models used in the study, and interpretation of the results. LMF contributed to the conception and design of the study, structure of the manuscript, and provided significant intellectual content and critical revision.

ABSTRACT

Background: Weight gain among young adults continues to increase. Identifying adults at high risk for weight gain and intervening before they gain weight could have a major public health impact. Our objective was to develop and test electronic health record-based (EHR) machine learning models to predict weight gain in young adults with overweight/class 1 obesity.

Materials and Methods: Seven machine learning models were assessed, including three regression models, random forest, single-layer neural network, gradient-boosted decision trees (XGBoost), and support vector machine (SVM) models. Four categories of predictors were included: 1) demographics; 2) obesity-related health conditions; 3) laboratory data and vital signs; and 4) neighborhood-level variables. The cohort was split 60:40 for model training and validation. Area under the receiver operating characteristic curves (AUC) were calculated to determine model accuracy at predicting high-risk individuals, defined by $\geq 10\%$ total body weight (TBW) gain within two years. Variable importance was measured via generalized ANOVA procedures.

Results: Of the 24,183 patients (mean [SD] age, 32.0 [6.3] years; 55.1% female) in the study, 14.2% gained $\geq 10\%$ TBW. AUCs varied from 0.557 (SVM) to 0.675 (XGBoost). Age, sex, and baseline BMI were the most important predictors among the models except SVM and neural network.

Conclusions: Our machine learning models performed similarly and had modest accuracy for identifying young adults at risk of weight gain. Future models may need to incorporate behavioral and/or genetic information to enhance model accuracy.

Keywords: Machine learning; adult, young; weight gain; obesity*

INTRODUCTION

Severe obesity, defined as a body mass index (BMI) ≥ 35 kg/m², affects 15% of the U.S. adult population.¹ From 2011 to 2018, the prevalence of severe obesity in the U.S. increased nearly 50% among young adults.^{2,3} Weight gain during early adulthood is associated with a decreased quality of life and an increased risk of developing obesity-related comorbidities and cancers.⁴⁻⁶ Medical costs directly attributable to obesity account for \$260.6 billion additional healthcare spending in the U.S., making prevention paramount.⁷

Few studies have focused on predicting which adults are at risk for weight gain, and none have used electronic health record (EHR) data to apply machine learning approaches to identify adults at risk for weight gain. Thus, we have been unable to target weight gain prevention efforts within health care systems. Machine learning models, which can process large amounts of data, can be used as clinical decision-making aids, and be incorporated into the EHR to generate real-time notifications. They have been applied to early childhood and adolescent obesity prediction,^{8,9} weight loss prediction,^{10,11} and the examination of environmental factors associated with obesity prevalence.^{12,13} Machine learning approaches have been incorporated into the EHR to predict acute kidney injury and trauma mortality.^{14,15}

The objective of this study was to develop machine learning models and test their accuracy at predicting which young adults are at the highest risk of weight gain using data available in the EHR. We hypothesized that available EHR variables would yield one or more models with an area under receiver operating characteristic curve (AUC) that exceeded a cutoff of 0.7, suggesting that the model was accurate.

METHODS

Study Setting

We used data from the University of Wisconsin Hospital and Clinics (UW) EHR from June 2008 to December 2018. Patients between 19 to 39 years old with a baseline BMI between 25.0-34.9 kg/m² and ≥ 2 weight measurements 2 years apart (± 6 months) were eligible for inclusion. The BMI range encompassing overweight and class 1 obesity was selected to target patients who were at risk of developing comorbidities with significant weight gain.¹⁶ Patients were required to have an encounter at least one year before the initial BMI measurement to ensure they were in the system long enough to obtain baseline data (e.g., vital signs, laboratory values, medical history) for the models. For patients with multiple 2-year intervals with BMI measurements from 2008-2019, a random interval was chosen to reduce selection bias.

Patients were excluded if they had a diagnosis of pregnancy within 12 months, prior bariatric surgery, or a cancer diagnosis (with exception of local squamous cell or basal cell skin cancer). Pregnancy and cancer diagnoses were identified using ICD-9 and -10 codes (see **eTable 1**). The UW bariatric surgery registry, which captures all bariatric surgery patients,¹⁷ was used to exclude patients with a history of bariatric surgery.

Primary Outcome

The primary outcome for all models was $\geq 10\%$ total body weight (TBW) gain within two years (± 6 months) of the baseline BMI measurement. The use of 10% TBW gain over 2 years was extrapolated from a combination of 5% weight loss being clinically significant,^{18,19} corresponding to the top 15-20% of the population for weight gain, and roughly equating to the amount and/or rate of weight gain associated with increased risk.^{4-6,20}

Predictors

Four categories of variables, which were all extracted from the EHR, were included in the models (see **eTable 2** for more detail):

1. **Patient demographics** - Demographic variables included baseline BMI, baseline age, sex, race and ethnicity (Asian non-Hispanic, Black non-Hispanic, Hispanic, Native American, White non-Hispanic, other), insurance type (commercial, Medicaid, Medicare, other), and smoking status (never, current, former). If insurance status was not available during the baseline encounter, it was obtained from the most recent previous encounter with non-missing insurance. Similar to our prior EHR studies, BMI data was “cleaned” using a methodology proposed by Cheng et al. to remove inaccurate and implausible heights and weights.^{16,21}
2. **Obesity-related health conditions** - anxiety, cardiovascular disease, chronic pain, depression, gastroesophageal reflux disease, hyperlipidemia, hypertension, obstructive sleep apnea, osteoarthritis, pre-diabetes, and Type 2 diabetes were identified using ICD-9 and -10 codes (see **eTable 1**). These conditions were selected due to their associations with overweight/obesity.¹⁶
3. **Laboratory data and vital signs** – Lab data included alanine transaminase (ALT), creatinine, glucose, HDL cholesterol, hemoglobin A1C, LDL cholesterol, and thyroid stimulating hormone (TSH). Given that there are no published criteria for how to manage outliers in the data, we referred to the CDC National Health and Nutrition Examination Survey (NHANES) data for laboratory value ranges and excluded values outside the range.²² We did not exclude variables with a high degree of missingness nor did we impute missing values given that the presence/absence of a value for that variable could be indicative of a patient’s utilization of the health care system and overall health. Instead, laboratory values were categorized as

normal, abnormal, or missing using the most recent value preceding the baseline BMI measurement (**eTable 2**). Vital signs included systolic and diastolic blood pressure and heart rate. Although there is not a standard cut-off, we excluded blood pressures of < 70/40 mmHg and heart rates < 40 as these are not physiologically compatible with outpatient encounters.

4. Neighborhood-level characteristics - Three characteristics at the neighborhood level (i.e., census block group level which is comprised of 600 to 3,000 people) were included: (1) area deprivation index (ADI),²³ which is a composite measure that describes neighborhood socioeconomic deprivation^{24,25}; (2) walkability index, which is a composite measure characterizing neighborhood walkability²⁶; (3) neighborhood urbanicity using the rural-urban commuting area (RUCA) database.²⁷ Patient census-block group was identified using the address at the time of baseline encounter. **eTable 2** contains more detailed variable explanations.

Model Development

The dataset was divided into training (60%) and validation (40%) cohorts, which is a commonly used approach in splitting datasets in machine learning studies.^{12,14,28} Stratified sampling was utilized to ensure each dataset had a similar proportion of events ($\geq 10\%$ TBW gain) to prevent the data from being unbalanced.²⁹ The training dataset was used for all model tuning using ten-fold cross-validation to maximize the training set AUC.³⁰ We evaluated seven predictive algorithms: logistic regression, ridge regression, lasso regression, random forest, single-layer neural network, gradient-boosted decision trees (XGBoost), and support vector machine (SVM). **Supplementary eMethods** and **eTable 3** includes additional information on model development and tuning.

Model Evaluation

The validation cohort was used to evaluate the performance of all models based on sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and accuracy for a range of predicted probability thresholds. The prediction threshold that maximized the sensitivity and specificity was selected. AUCs were calculated to determine the models' overall accuracy at predicting which individuals would gain $\geq 10\%$ TBW. Standard deviations were obtained using the DeLong method.³¹

To assess variable importance, we used a model-agnostic approach that utilizes a nonparametric generalization of the classical ANOVA-based measure of variable importance.³² We refer to this measure as the Williamson variable importance measure (WVIM). For each variable, WVIM calculates the improvement in predictive performance using the full set of covariates relative to using all the covariates *except* the variable of interest. Because the measure is based on a nonparametric formulation, it is model-agnostic, allowing WVIM to be used for all the models in our study. To investigate the top three most influential variables across all models, we conducted two *post hoc* model analyses to evaluate the change in AUC for prediction $\geq 10\%$ TBW gain: (1) with only the top three most influential variables, and (2) with all variables other than the top three most important variables.

Analyses were performed using R version 4.0.2 (The R Foundation for Statistical Computing 2020; Vienna, Austria) and SAS version 9.4 (SAS Institute; Cary, North Carolina, U.S.). The UW-Madison Minimal Risk Institutional Review Board (IRB) approved the study (#2019-0190). The need for informed consent was waived. The Strengthening the Reporting of Observational Studies in Epidemiology guidelines were followed in the reporting and methodology of this study.³³

RESULTS

Study Cohort Characteristics

24,183 patients were included in the final cohort: 3,428 patients who had gained $\geq 10\%$ TBW and 20,755 who had $< 10\%$ TBW gain (**Figure 1**). When compared to patients who did not gain $\geq 10\%$ TBW, patients who gained $\geq 10\%$ TBW were younger (mean age 30.3 [SD 6.6] versus 32.3 [6.2], p-value < 0.0001), more likely to be female (69.3% vs 52.8%, p-value < 0.0001), more likely have to have a diagnosis of anxiety (37.1% vs 28.6%, p-value < 0.0001) or chronic pain (10% vs 7.7%, p-value < 0.0001), and more likely to have Class 1 obesity at baseline (50% vs 34.5%, p < 0.001 ; **Table 1**).

Missingness was lowest among the following categories of variables: census block group variables (0.04%), baseline demographic variables (age, sex [0.00%]; race and ethnicity [0.8%]; and insurance status [0.6%]) and vital signs (heart rate [3.7%] and blood pressure [0.7%]). Laboratory test results had the most missingness, ranging from glucose (74.2%) to hemoglobin A1C (96.1%) (**Table 1**). Vital sign and laboratory test distributions are displayed in **eTable 4**.

Performance Evaluation

In the validation dataset, XGBoost has the highest discrimination (AUC 0.675 [95% CI 0.660-0.690]), while SVM (AUC 0.557 [95% CI 0.541-0.573]) and random forest (AUC 0.649 [95% CI 0.632-0.663]) had the lowest discrimination (**Figure 2**). **Table 2** displays the additional parameters of sensitivity, specificity, PPV, NPV, and accuracy for the models at a predicted probability threshold of 0.2. Models had higher specificity and NPV, but lower sensitivity and PPV. **eTable 5** displays the additional model parameters corresponding to predictive probability thresholds ranging from 0.1 to 0.5.

Variable Importance

Baseline BMI, age, and sex were the most important predictors for all models except neural network and SVM. Area deprivation index, diabetes, and cardiovascular disease were the most important for neural network. Age, baseline BMI, and thyroid stimulating hormone were the most important variables for SVM (**Figure 3**).

Additional Analyses

In *post hoc* analyses of models that included only the three most important predictors (baseline BMI, age, and sex), only random forest had a meaningful decrease in AUC when compared to a model that included the full variable set. Random forest with only baseline BMI, age, and sex had a decrease in AUC from 0.648 (95% CI 0.633, 0.664) to 0.588 (95% CI 0.572, 0.604). All models performed significantly worse with the removal of baseline BMI, age, and sex with AUCs ranging from 0.490 (95% CI 0.473, 0.507) for SVM to 0.588 (95% CI 0.571, 0.604) for lasso, ridge, and logistic regression (**eFigure 1**).

DISCUSSION

In this large observational analysis, we found that machine learning methods performed similarly and had modest accuracy for identifying young adults at risk for gaining a clinically significant amount of weight using existing EHR data. Age, baseline BMI, and sex were the most important variables for most models. Model performance may have been hampered by high degree of missingness in the laboratory variables.

Contrary to our hypothesis, the machine learning models did not accurately predict who would gain a clinically significant amount of weight above our pre-defined threshold of an AUC of 0.70. Furthermore, the four “modern” machine learning models (random forest, XGBoost,

neural network, and SVM) did not significantly outperform traditional regression models. This lack of prediction capability using electronic health record data – both in our models and in the literature - represents an opportunity to improve preventative care of a major public health challenge. Application of accurate machine learning algorithms using EHR data has been successful in other fields, such as prediction of sepsis risk,²⁸ need for adjuvant cancer therapy,³⁴ cirrhosis mortality,³⁵ diabetes onset,^{36,37} and cardiovascular disease risk.^{38,39} These models differ from ours by predicting outcomes in an inpatient population or focusing on conditions with regular screening imaging/labs in the outpatient setting.^{14,34-39}

Models incorporating a more informative set of predictors, including individual lifestyle habits like dietary habits and physical activity, and socioeconomic factors such as food insecurity, should be evaluated. Survey-based and retrospective cohort studies have found that obesity is positively associated with the amount of television that an individual watches per day, consumption of meals prepared away from home, and less than a college education.⁴⁰⁻⁴² Two datasets that have been introduced within the past decade have significant potential for obesity prediction modeling: the National Institutes of Health *All of Us* Research Program (*All of Us*) with nearly 500,000 individuals and the Veteran's Affairs Million Veteran Program (MVP) with over one million Veterans.⁴³ In addition to EHR data, both datasets contain lifestyle and socioeconomic surveys along with genetic and laboratory data. The behavioral components of these datasets would allow for the assessment of self-reported lifestyle habits (e.g., diet, exercise, screen time) that may be more predictive of weight gain than variables currently reported in the EHR.^{41,44} Incorporation of these variable types into the EHR could allow for their utilization in a wide range of predictive modeling as machine learning utilization becomes more prevalent.

Age, sex, and baseline BMI were the most important predictors for all models except SVM and neural network. This is consistent with other published literature which has found them to be associated with weight gain. Age was the most important variable in a Finnish database study on predictors associated with obesity using random forest and logistic regression models.⁴⁵ Weight has been shown to be a strong predictor for future obesity in machine learning models focused on children. Dugan *et al.* developed a tree-based model with 85% accuracy at predicting a child's obesity risk as an adult. The most important predictor was meeting criteria for overweight before two years of age.⁹ Similarly, Singh *et al.* used BMI trend from age 3 to 11 to predict the risk of having obesity or overweight as an adult.⁴⁶ Notably, studies that have used machine learning methods to predict childhood weight gain or weight loss outcomes following medical or surgical intervention have relied on longitudinal BMI trends with multiple time points for data entry.^{9,11,47,48} Our model incorporated a single baseline BMI value. Future model evaluation should consider incorporation of BMI trends, or trends in other variables (e.g., glucose values), as these may prove to be more predictive. However, requiring multiple data points for study inclusion would decrease the sample size and delay initial model prognostication until the patient has had multiple interactions with the healthcare system.

The neighborhood-level variables were not drivers of model prediction in our study, despite prior studies linking ADI and walkability with obesity risk.^{42,49,50} Other machine learning models have attributed high variable importance to county-level socioeconomic markers (median household income and percent with college education) in predicting obesity rates.¹³ The relative homogeneity of our population at the neighborhood-level variables (e.g., low percentage of high ADI or rural patients) may have decreased the variable importance as the models were not able to identify patterns due to the limited variability. Training models with a cohort of patients with

more neighborhood-level variation may improve prognostication. Another possibility is that individual behaviors are more predictive than the neighborhood environment. This is supported by Creatore *et al.* who reported that although obesity and walkability were correlated, the amount of physical activity an individual reported was not correlated with neighborhood walkability.⁴⁹

Our study has limitations. First, there may be selection bias caused by the requirement of two weight measurements within two years. Second, young adults with multiple encounters in the healthcare system are likely not representative of the entire young adult population. However, the goal is to identify patients who are being seen in the healthcare system to allow for early intervention. Third, the laboratory variables had a high degree of missing data, although missingness was quite low for the most important model variables (e.g., age). Fourth, our study population was from a single health care system which may impede generalizability to other settings. Fifth, there may be data entry errors into the EHR that impacted our findings. We “cleaned” the BMI, laboratory, and vital sign data prior to input into the models, but inaccuracies in the data may still exist. Sixth, there may be observational bias due to unmeasured confounding. Finally, there was class imbalance in the outcome measure. We considered approaches to address this imbalance, such as downsampling the majority group or upsampling the minority group. However, we did not perform these analyses because in our experience they typically do not improve performance when the outcome prevalence is >10%, as in our study.⁵¹

CONCLUSIONS

We found that machine learning approaches to predict weight gain using a single institution’s electronic health record data performed similarly and were relatively inaccurate. Model optimization using more informative datasets is warranted. If model classification

accuracy improves, incorporating machine learning models into clinical care may allow for early identification of at-risk young adults and referral to weight management programs.

DISCLOSURES

Effort on this study and manuscript were made possible by an NIH R-21 (R21MD012655-01), an American College of Surgeons George H.A. Clowes Career Development Award, and a VA Career Development Award (CDA 015-060) to Dr. Funk. Further funding was through the NIH Metabolism and Nutrition Training Program T32 (DK 007665) to Dr. Murtha. Dr. Churpek is supported by R01s from NIGMS (R01 GM123193), NHLBI (R01 HL157262), NIDDK (R01-DK126933), and a PRMRP grant from DOD (W81XWH-21-1-0009). He also has a patent pending (ARCD. P0535US.P2) for risk stratification algorithms for hospitalized patients.

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FIGURE TITLES AND LEGENDS

Table 1. Variables included in regression analysis with summary statistics

p-value compares cohort that gained $\geq 10\%$ TBW to cohort that did not gain $\geq 10\%$ TBW

^a Denotes categories of data with no missing variables.

^b Missing race and ethnicity n=204 (0.8%); Other includes other, Asian, Native American

^c Missing insurance n=134 (0.6%); Other included fee for service, workers compensation

^d Missing smoking status n=113 (0.5%)

^e Missing zip code n=11 (0.00%)

Table 2. Model Evaluation and Validation

Evaluation and validation are at a predicted probability threshold of 0.2. AUC: area under the receiver operator characteristic curve; PPV: positive predictive value; NPV: negative predictive value.

Figure 1. Cohort selection

Flowchart of study cohort selection

Figure 2. Area under receiver operating characteristic curves of the compared models for prediction of weight gain risk

Area under the receiver operator characteristic curves (AUCs) of the compared models for prediction of $\geq 10\%$ total body weight gain with 95% DeLong confidence intervals.

Figure 3. Variable importance plot

All model variables are listed on y-axis. Model types are displayed on x-axis. Darker shading correlates with higher variable importance.

TABLES

Table 1. Variables included in regression analysis with summary statistics

	Entire cohort n=24,183	Did not gain ≥10% TBW n=20,755	Gained ≥10% TBW n=3,428	p-value
Age, mean (SD), yrs ^a	32.0 (±6.3)	32.3 (±6.2)	30.3 (±6.6)	< 0.0001
Sex (n,%) ^a				
Male	10,856 (44.9)	9,802 (47.2)	1,054 (30.8)	< 0.0001
Female	13,327 (55.1)	10,953 (52.8)	2,374 (69.3)	
Race and ethnicity (n,%) ^b				
White, non-Hispanic	20,468 (84.6)	17,554 (84.6)	2,914 (85.0)	< 0.0001
Black	1,429 (5.9)	1,161 (5.6)	268 (7.8)	
Hispanic	970 (4.0)	848 (4.1)	122 (3.6)	
Other/unspecified	1,112 (4.6)	1,017 (4.9)	95 (3.3)	
Insurance type (n,%) ^c				
Commercial	21,734 (89.9)	18,779 (90.5)	2,955 (86.2)	< 0.0001
Medicare	811 (3.4)	642 (3.1)	169 (0.7)	
Medicaid	848 (3.5)	664 (3.2)	184 (5.4)	
Other	656 (2.7)	562 (2.7)	94 (2.7)	
Baseline BMI category (n,%) ^a				
Overweight (25.0–29.9 kg/m ²)	15,310 (63.3)	13,595 (65.5)	1,715 (50.0)	< 0.0001
Class 1 obesity (30.0–34.9 kg/m ²)	8,873 (36.7)	7,160 (34.5)	1,713 (50.0)	
Smoking status (n,%) ^d				
Never smoker	14,451 (59.8)	12,537 (60.4)	1,914 (55.8)	< 0.0001
Former/current smoker	9,619 (39.8)	8,124 (39.1)	1,495 (43.6)	
Unknown	113 (0.5)	94 (0.5)	19 (0.6)	
Health conditions (n,%)				
Anxiety	7,198 (29.8)	5,928 (28.6)	1,270 (37.1)	< 0.0001
Gastroesophageal reflux disease	4,063 (16.8)	3,470 (16.7)	593 (17.3)	0.4019
Hyperlipidemia	3,922 (16.2)	3,395 (16.4)	527 (15.4)	0.1540
Hypertension	2,585 (10.7)	2,175 (10.5)	410 (12.0)	0.0103
Chronic pain	1,934 (8.0)	1,590 (7.7)	344 (10.0)	< 0.0001
Obstructive sleep apnea	1,176 (4.9)	979 (4.7)	197 (5.8)	0.0114
Pre-diabetes	1,163 (4.8)	964 (4.6)	199 (5.8)	0.0039
Depression	1,077 (4.5)	850 (4.1)	227 (6.6)	< 0.0001

Type 2 diabetes	766 (3.2)	651 (3.1)	115 (3.4)	0.4939
Osteoarthritis	676 (2.8)	585 (2.8)	91 (2.7)	0.6151
Cardiovascular disease	116 (0.5)	97 (0.5)	19 (0.6)	0.5038
Vital signs (n,%)				
Heart rate				0.1749
Bradycardia	1,004 (4.2)	866 (4.2)	138 (4.0)	
Normal	21,188 (87.6)	18,203 (87.7)	2,985 (87.1)	
Tachycardia	1,088 (4.5)	909 (4.4)	179 (5.2)	
Missing	903 (3.7)	777 (3.7)	126 (3.7)	
Blood pressure				0.0166
Normal	15,177 (62.8)	12,954 (62.4)	2,223 (64.9)	
Pre-hypertension	6,504 (26.9)	5,614 (27.1)	890 (26.0)	
Hypertension	2,345 (9.7)	2,054 (9.9)	291 (8.5)	
Missing	157 (0.65)	133 (0.6)	24 (0.7)	
Laboratory Test Results (n, %)				
Hemoglobin A1C				0.0044
Normal	456 (1.9)	369 (1.8)	87 (2.5)	
Pre-diabetes	171 (0.7)	138 (0.7)	33 (1.0)	
Diabetes	319 (1.3)	276 (1.3)	43 (1.3)	
Missing	23,237 (96.1)	19,972 (96.2)	3,265 (95.5)	
LDL cholesterol				<0.0001
Normal	1,280 (5.3)	1,055 (5.1)	225 (6.6)	
Borderline	374 (1.6)	336 (1.6)	38 (1.1)	
Abnormal	164 (0.7)	153 (0.7)	11 (0.3)	
Missing	22,365 (92.5)	19,211 (92.6)	3,154 (92.0)	
Creatinine				<0.0001
Normal	5,510 (22.8)	4,551 (21.9)	959 (28.0)	
Abnormal	456 (1.9)	385 (1.9)	71 (2.1)	
Missing	18,217 (75.3)	15,819 (76.2)	2,398 (70.0)	
Glucose				<0.0001
Normal	4,366 (18.1)	3,629 (17.5)	737 (21.5)	
Pre-diabetes	1,315 (5.4)	1,098 (5.3)	217 (6.33)	
Diabetes	570 (2.4)	467 (2.3)	103 (3.0)	
Missing	17,932 (74.2)	15,561 (75.0)	2,371 (69.2)	
HDL cholesterol				0.4476
Normal	457 (1.9)	383 (1.9)	74 (2.2)	
Abnormal	2,070 (8.6)	1,774 (8.6)	296 (8.6)	
Missing	21,656 (89.6)	18,598 (89.6)	3,058 (89.2)	
TSH				<0.0001
Normal	3,794 (15.7)	3,148 (15.2)	646 (18.8)	
Hypothyroid	292 (1.2)	252 (1.2)	40 (1.2)	
Hyperthyroid	83 (0.3)	61 (0.3)	22 (0.6)	
Missing	20,014 (82.8)	17,294 (83.3)	2,720 (79.4)	
Alanine transaminase				<0.0001
Normal	3,806 (15.7)	3,139 (15.1)	667 (19.5)	
Abnormal	535 (2.2)	468 (2.3)	67 (2.0)	
Missing	19,842 (82.1)	17,148 (82.6)	2,694 (78.6)	
Census Block Group Variables ^e				
Area deprivation index (mean, SD)	36.7 (±18.5)	36.3 (±18.3)	39.3 (±19.1)	< 0.0001

Walkability Index (mean, SD)	9.1 (±4.1)	9.1 (±4.1)	9.1 (±4.1)	0.9016
Urbanicity (n,%)				0.0001
<i>Urban</i>	20,940 (86.6)	18,049 (87.0)	2,891 (84.3)	
<i>Suburban</i>	1,610 (6.7)	1,351 (6.5)	259 (7.6)	
<i>Rural</i>	1,633 (6.8)	1,355 (6.5)	278 (8.1)	
p-value compares cohort that gained ≥10% TBW to cohort that did not gain ≥10%TBW				
^a Denotes categories of data with no missing variables.				
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^d Missing smoking status n=113 (0.5%)				
^e Missing zip code n=11 (0.00%)				

Table 2. Model Evaluation and Validation

Model	AUC	Sensitivity	Specificity	PPV	NPV	Accuracy
Logistic regression (linear)						
Development cohort	0.691	0.409	0.823	0.276	0.894	0.764
Validation cohort	0.669	0.378	0.823	0.261	0.889	0.760
Logistic regression (spline)						
Development cohort	0.691	0.415	0.823	0.279	0.895	0.765
Validation cohort	0.669	0.381	0.818	0.256	0.889	0.756
Lasso regression						
Development cohort	0.699	0.416	0.830	0.288	0.896	0.771
Validation cohort	0.671	0.374	0.829	0.267	0.889	0.764
Ridge regression						
Development cohort	0.701	0.431	0.826	0.291	0.898	0.770
Validation cohort	0.670	0.369	0.824	0.257	0.888	0.760
Random forest						
Development cohort	0.652	0.465	0.731	0.222	0.892	0.693
Validation cohort	0.649	0.407	0.780	0.234	0.889	0.727
Neural net						
Development cohort	0.699	0.193	0.550	0.722	0.101	0.244
Validation cohort	0.669	0.400	0.806	0.256	0.891	0.749
Support vector machine						
Development cohort	0.972	0.819	0.999	0.971	0.996	0.974
Validation cohort	0.557	0.025	0.979	0.859	0.163	0.844
XGBoost						
Development cohort	0.721	0.442	0.834	0.306	0.901	0.779
Validation cohort	0.675	0.368	0.827	0.259	0.888	0.762
Evaluation and validation are at a predicted probability threshold of 0.2. AUC: area under receiver operating characteristic curve; PPV: positive predictive value; NPV: negative predictive value.						

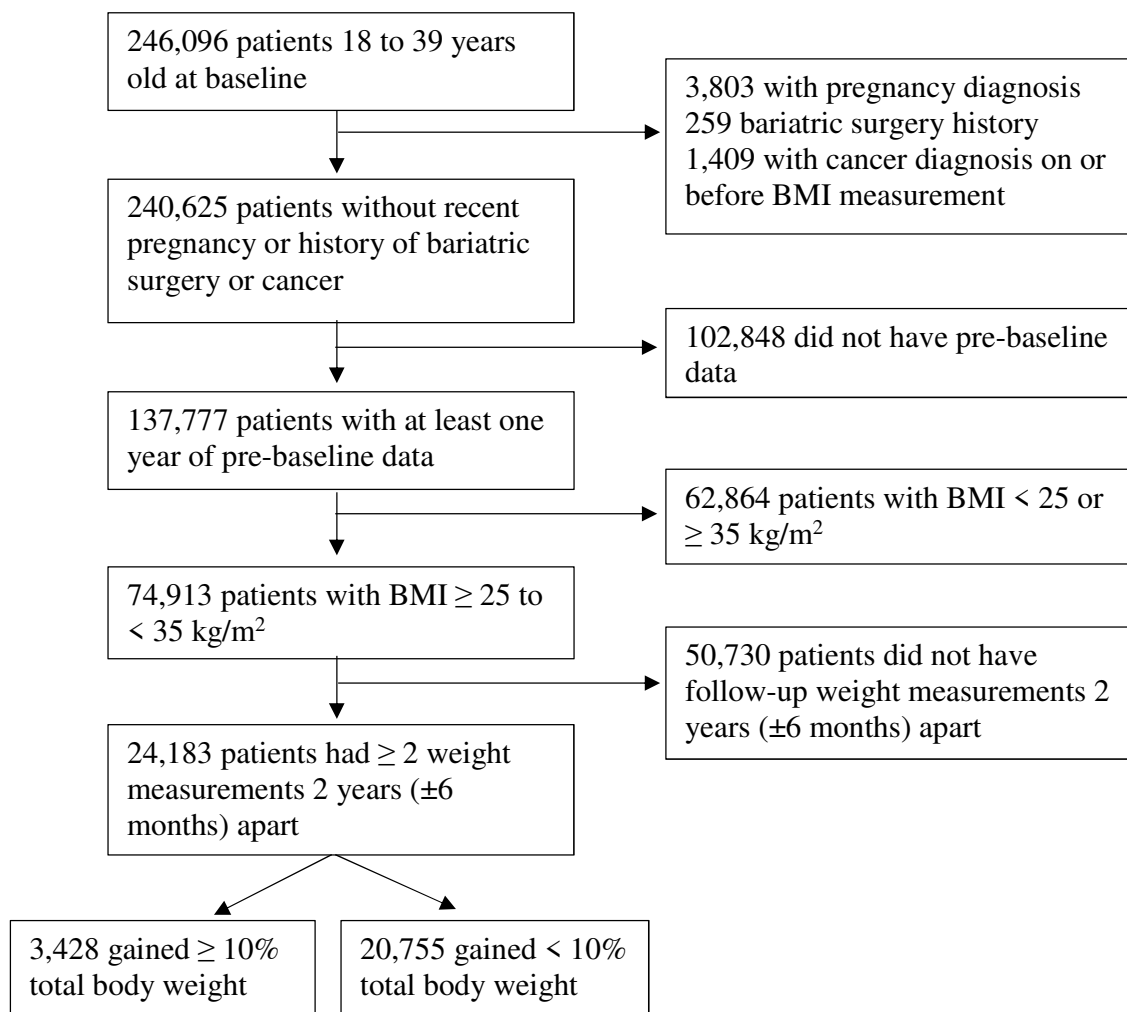
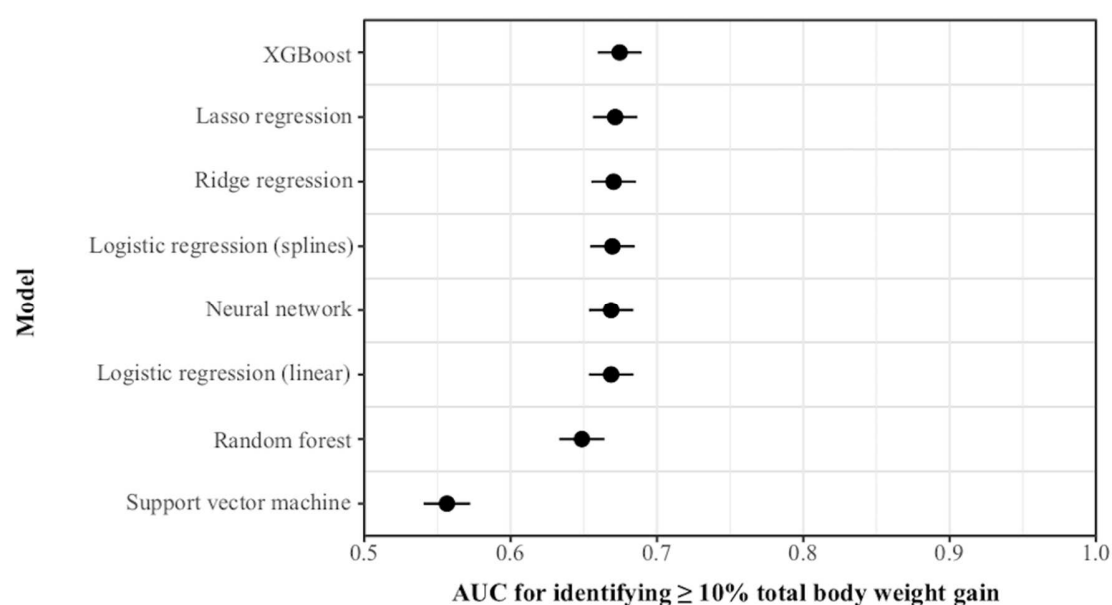
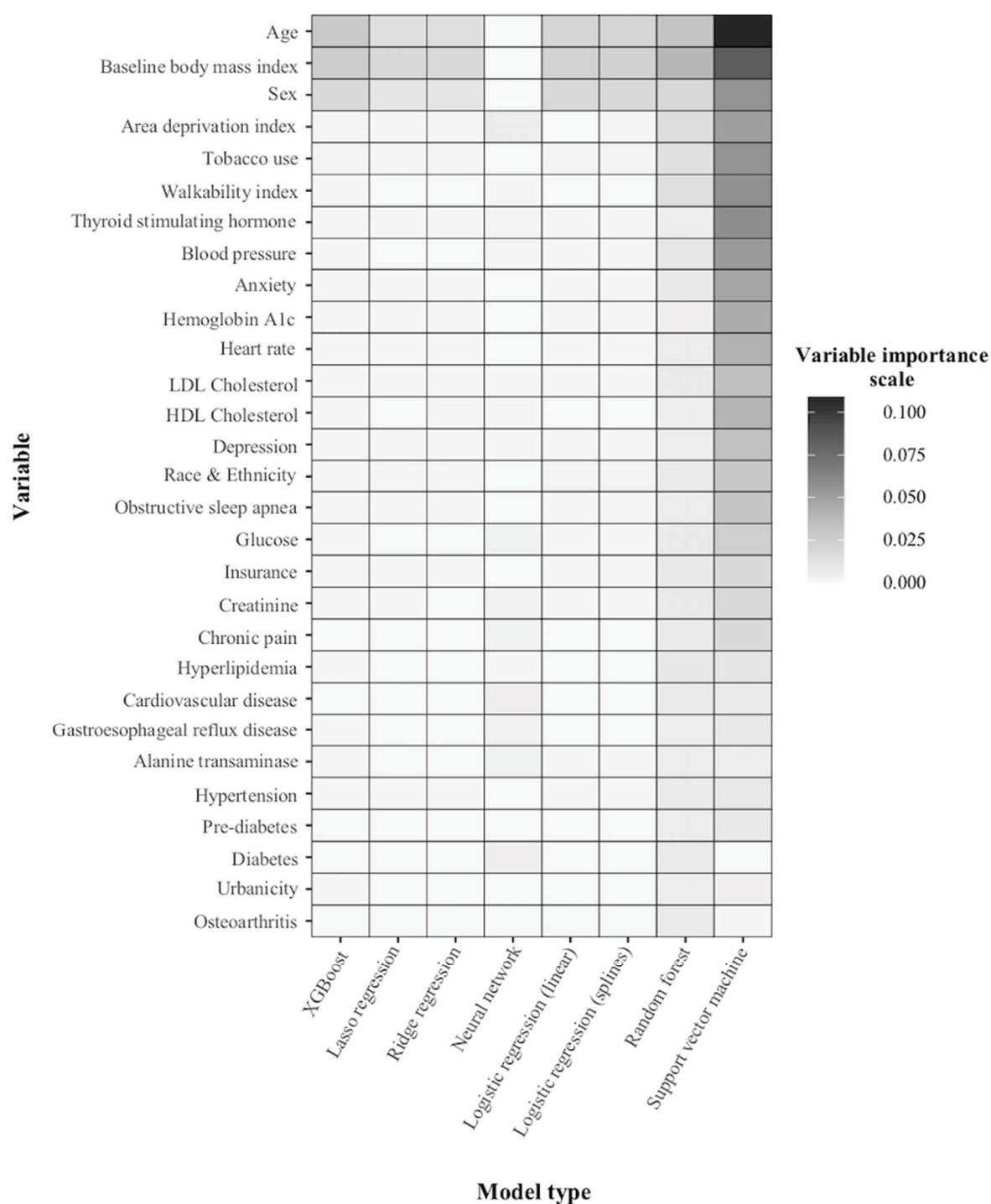
Figure 1. Cohort selection

Figure 2. Area under receiver operating characteristic curves of the compared methods for prediction of weight gain risk



Area under the receiver operator characteristic curves (AUCs) of the compared methods for prediction of $\geq 10\%$ total body weight gain with 95% DeLong confidence intervals

Figure 3. Variable importance plot



All model variables are listed on y-axis. Model types are displayed on x-axis. Darker shading correlates with higher variable importance.

SUPPLEMENTAL DIGITAL CONTENT

eMethods. Model overview and description of tuning parameters

eTable 1. ICD-9 and -10 codes used for exclusion criteria and comorbidities

eTable 2. Definition of variables included in machine learning algorithms

eTable 3. Machine learning models with considered parameters and top performing parameters

eTable 4. Vital sign and laboratory data distribution

eTable 5. Model parameters for range of predictive probability thresholds

eFigure 1. Sensitivity analyses with and without age, sex, and body mass index